

Configuration and provenance

Branch: feat/endtop-ssl4

Commits: 1f6aca1, 9613fc9, final EXEC_09 deliverable commit

- ▶ 31 positions $\times$ 2000 events; 16 End + 70 Top channels; jitter fixed to zero.
- ▶ EJ-204 via SSLG4 OPSC-101: yield 10400/MeV, $\tau_r = 0.7$ ns, $\tau_d = 1.8$ ns, ABSLENGTH=160 cm (verified effective MPT inputs).
- ▶ Broadcom AFBR-S4N66P024M: PDE 63% at 420 nm.

Scope

Top 70 SiPM is a simulation coverage study. Real hardware: 32 Top SiPMs. All reported timing is intrinsic, without SPTR or electronics jitter.

EndTop geometry and channel map

- ▶ EJ-204 bar: $1400 \times 60 \times 10 \text{ mm}^3$.
- ▶ End L IDs 0–7; End R IDs 8–15.
- ▶ Top L IDs 16–50: $-692, -672, \dots, -12 \text{ mm}$.
- ▶ Top R IDs 51–85: $+12, +32, \dots, +692 \text{ mm}$.
- ▶ Each Top element and window: $6 \times 6 \text{ mm}^2$.

Exact constraints

First centers are 5 mm from each bar end. Pitch is 20 mm except the deliberate central $-12/+12 \text{ mm}$ pair, whose pitch is 24 mm.

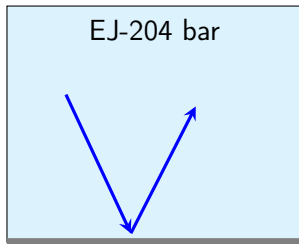
Instrumented and wrapped faces

Face	Optical state	Readout
$-X, +X$	open	8 End SiPMs each
$+Y$	Mylar with 70 exact windows	70 Top SiPMs
$-Y, -Z, +Z$	complete Mylar	none

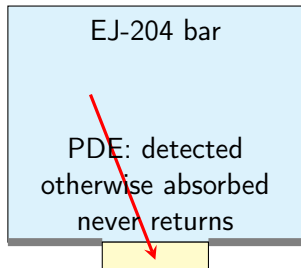
External layer

Black Tedlar is deliberately not modeled; it only blocks ambient light. The non-reflected Mylar fraction is already absorbed in the optical model.

Window optics: why timing and collection change



Mylar, fixed $R = 0.98$: reflected photons return

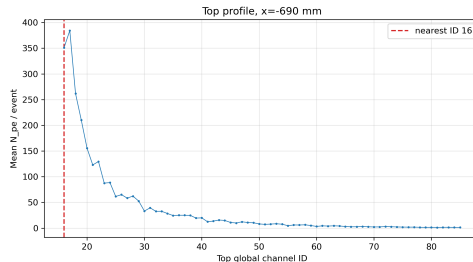
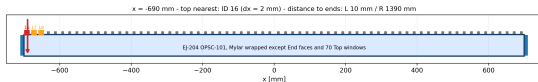


index-matched window

Consequence

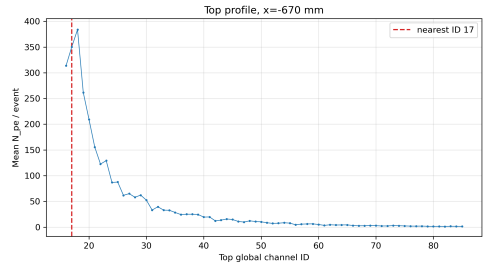
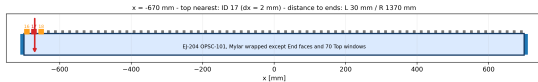
Every window is an optical drain. Long, multiply reflected paths have more chances to meet one, so late End photons are preferentially removed.

Position $x=-690$ mm



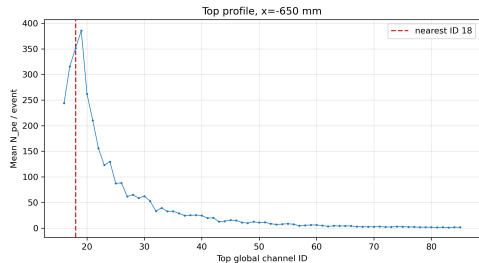
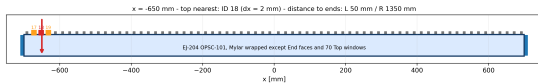
Quantity	Value
End L N_{pe}	3363.1 ± 18.1
End R N_{pe}	41.1 ± 0.3
Nearest Top ID	16 (exact geometry)
Nearest Top N_{pe}	350.9 ± 2.0
Nearest Top $\langle t \rangle$	2.786 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=-670$ mm



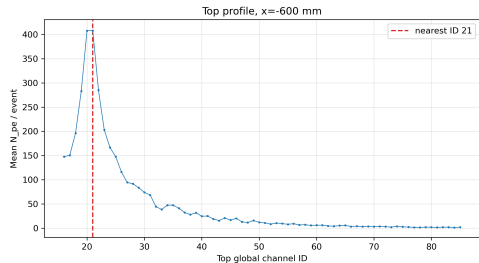
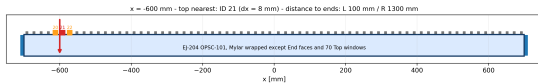
Quantity	Value
End L N_{pe}	2764.3 ± 15.0
End R N_{pe}	42.6 ± 0.3
Nearest Top ID	17 (exact geometry)
Nearest Top N_{pe}	350.4 ± 2.0
Nearest Top $\langle t \rangle$	2.775 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=-650$ mm



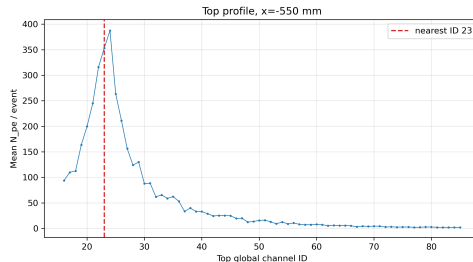
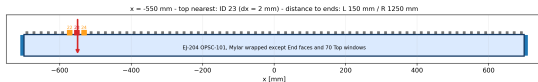
Quantity	Value
End L N_{pe}	2366.2 ± 13.1
End R N_{pe}	44.3 ± 0.3
Nearest Top ID	18 (exact geometry)
Nearest Top N_{pe}	351.0 ± 2.0
Nearest Top $\langle t \rangle$	2.778 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=-600$ mm



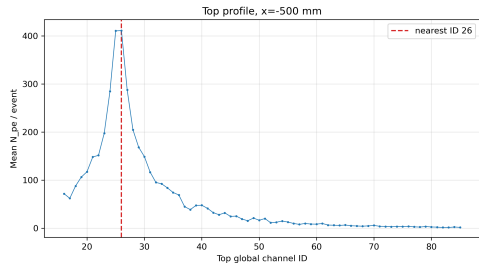
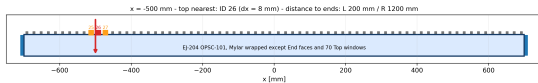
Quantity	Value
End L N_{pe}	1673.6 ± 9.2
End R N_{pe}	48.0 ± 0.3
Nearest Top ID	21 (exact geometry)
Nearest Top N_{pe}	408.0 ± 2.2
Nearest Top $\langle t \rangle$	2.942 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-550$ mm



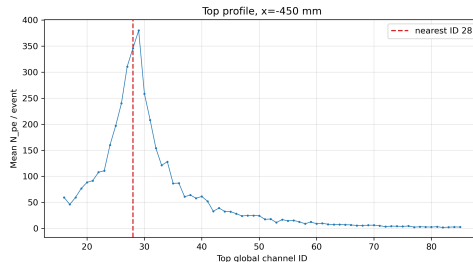
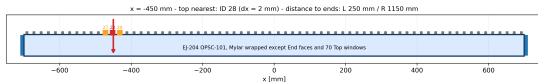
Quantity	Value
End L N_{pe}	1323.9 ± 7.6
End R N_{pe}	55.1 ± 0.4
Nearest Top ID	23 (exact geometry)
Nearest Top N_{pe}	352.9 ± 2.1
Nearest Top $\langle t \rangle$	2.776 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=-500$ mm



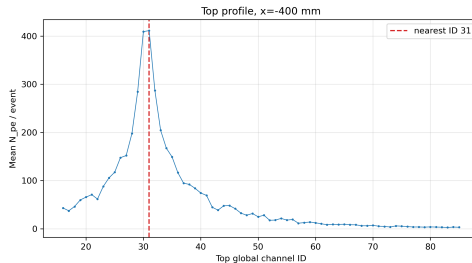
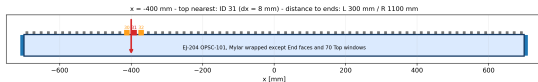
Quantity	Value
End L N_{pe}	1042.2 ± 5.9
End R N_{pe}	59.8 ± 0.4
Nearest Top ID	26 (exact geometry)
Nearest Top N_{pe}	410.9 ± 2.3
Nearest Top $\langle t \rangle$	2.942 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-450$ mm



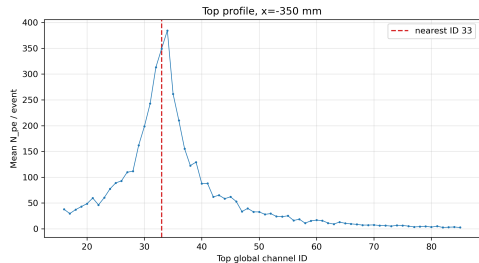
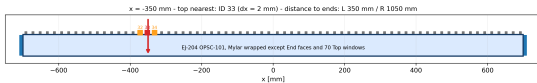
Quantity	Value
End L N_{pe}	832.9 ± 4.2
End R N_{pe}	65.8 ± 0.4
Nearest Top ID	28 (exact geometry)
Nearest Top N_{pe}	345.8 ± 1.8
Nearest Top $\langle t \rangle$	2.779 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=-400$ mm



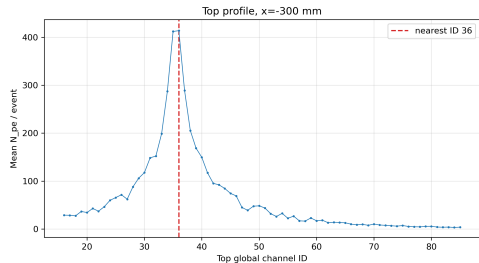
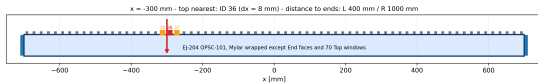
Quantity	Value
End L N_{pe}	691.3 ± 3.9
End R N_{pe}	74.5 ± 0.5
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	411.3 ± 2.3
Nearest Top $\langle t \rangle$	2.940 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-350$ mm



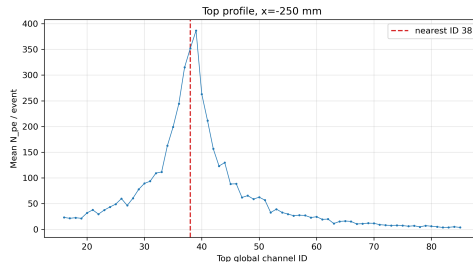
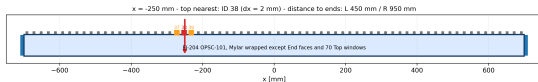
Quantity	Value
End L N_{pe}	575.0 ± 3.1
End R N_{pe}	84.1 ± 0.5
Nearest Top ID	33 (exact geometry)
Nearest Top N_{pe}	349.3 ± 1.9
Nearest Top $\langle t \rangle$	2.775 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=-300$ mm



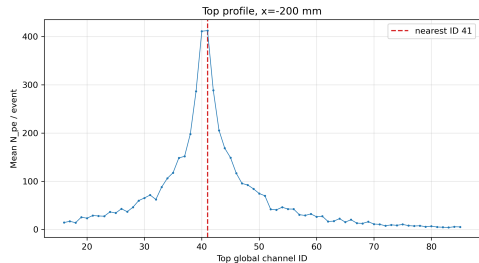
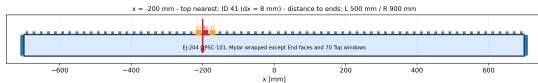
Quantity	Value
End L N_{pe}	482.3 ± 2.8
End R N_{pe}	94.2 ± 0.6
Nearest Top ID	36 (exact geometry)
Nearest Top N_{pe}	413.7 ± 2.5
Nearest Top $\langle t \rangle$	2.939 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-250$ mm



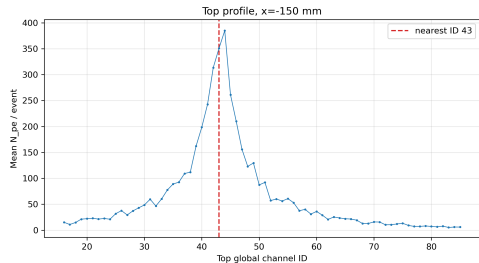
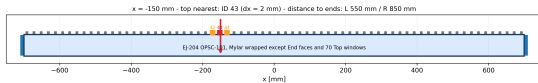
Quantity	Value
End L N_{pe}	415.8 ± 2.4
End R N_{pe}	105.5 ± 0.7
Nearest Top ID	38 (exact geometry)
Nearest Top N_{pe}	352.2 ± 2.1
Nearest Top $\langle t \rangle$	2.776 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=-200$ mm



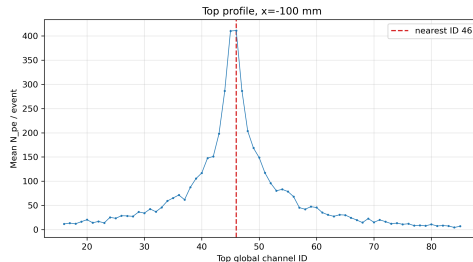
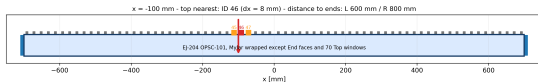
Quantity	Value
End L N_{pe}	353.5 ± 2.0
End R N_{pe}	118.4 ± 0.7
Nearest Top ID	41 (exact geometry)
Nearest Top N_{pe}	412.5 ± 2.3
Nearest Top $\langle t \rangle$	2.941 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-150$ mm



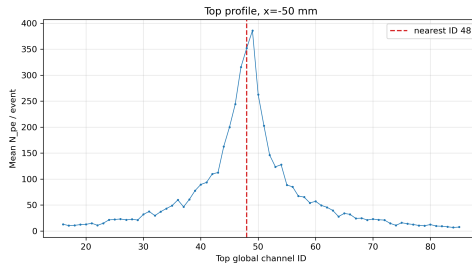
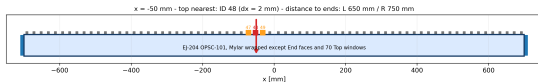
Quantity	Value
End L N_{pe}	302.1 ± 1.8
End R N_{pe}	134.7 ± 0.8
Nearest Top ID	43 (exact geometry)
Nearest Top N_{pe}	350.8 ± 2.1
Nearest Top $\langle t \rangle$	2.775 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=-100$ mm



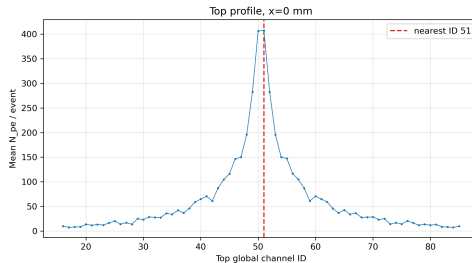
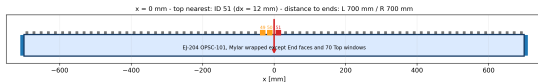
Quantity	Value
End L N_{pe}	256.3 ± 1.5
End R N_{pe}	151.4 ± 0.9
Nearest Top ID	46 (exact geometry)
Nearest Top N_{pe}	411.0 ± 2.3
Nearest Top $\langle t \rangle$	2.943 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=-50$ mm



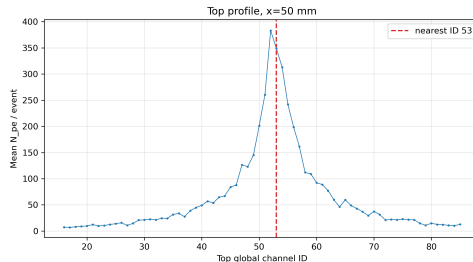
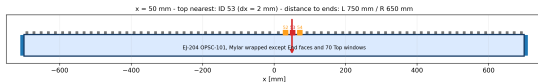
Quantity	Value
End L N_{pe}	226.7 ± 1.4
End R N_{pe}	172.7 ± 1.1
Nearest Top ID	48 (exact geometry)
Nearest Top N_{pe}	352.0 ± 2.1
Nearest Top $\langle t \rangle$	2.774 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=0$ mm



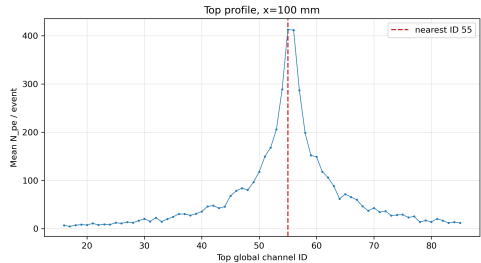
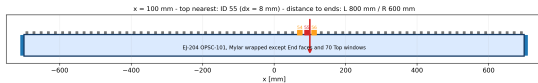
Quantity	Value
End L N_{pe}	194.4 ± 1.1
End R N_{pe}	195.2 ± 1.1
Nearest Top ID	51 (exact geometry)
Nearest Top N_{pe}	407.3 ± 2.2
Nearest Top $\langle t \rangle$	3.003 ± 0.002 ns
Nearest Top FPT	0.351 ± 0.000 ns

Position $x=50$ mm



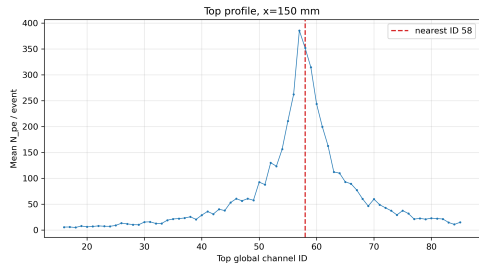
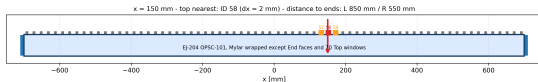
Quantity	Value
End L N_{pe}	171.6 ± 1.1
End R N_{pe}	226.4 ± 1.3
Nearest Top ID	53 (exact geometry)
Nearest Top N_{pe}	348.8 ± 2.1
Nearest Top $\langle t \rangle$	2.779 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=100$ mm



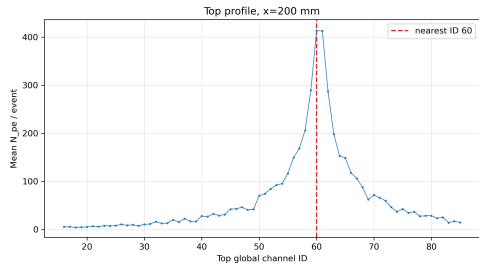
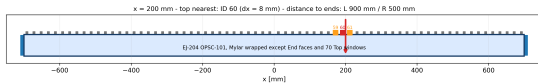
Quantity	Value
End L N_{pe}	152.2 ± 0.8
End R N_{pe}	258.3 ± 1.4
Nearest Top ID	55 (exact geometry)
Nearest Top N_{pe}	412.8 ± 2.2
Nearest Top $\langle t \rangle$	2.943 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=150$ mm



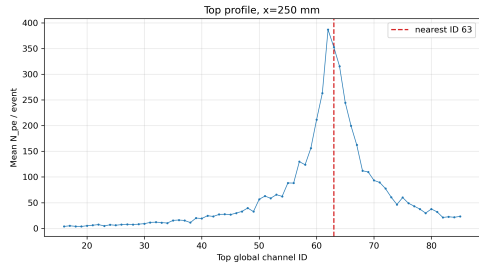
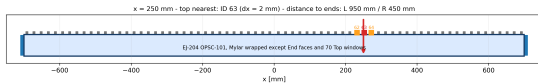
Quantity	Value
End L N_{pe}	135.5 ± 0.8
End R N_{pe}	303.6 ± 1.7
Nearest Top ID	58 (exact geometry)
Nearest Top N_{pe}	352.1 ± 2.0
Nearest Top $\langle t \rangle$	2.782 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=200$ mm



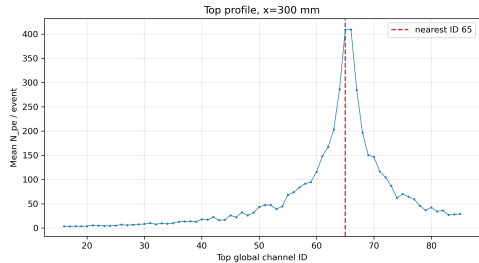
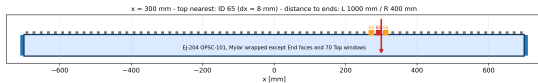
Quantity	Value
End L N_{pe}	118.5 ± 0.7
End R N_{pe}	354.1 ± 2.1
Nearest Top ID	60 (exact geometry)
Nearest Top N_{pe}	413.3 ± 2.4
Nearest Top $\langle t \rangle$	2.937 ± 0.002 ns
Nearest Top FPT	0.345 ± 0.000 ns

Position $x=250$ mm



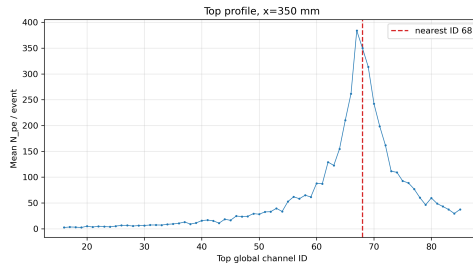
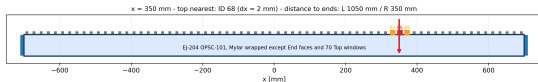
Quantity	Value
End L N_{pe}	104.8 ± 0.7
End R N_{pe}	417.0 ± 2.4
Nearest Top ID	63 (exact geometry)
Nearest Top N_{pe}	353.0 ± 2.1
Nearest Top $\langle t \rangle$	2.776 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=300$ mm



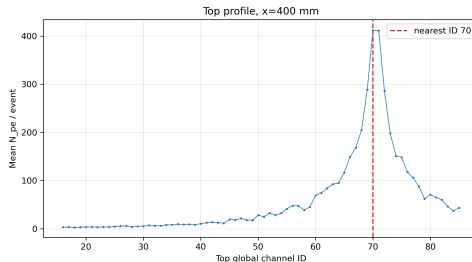
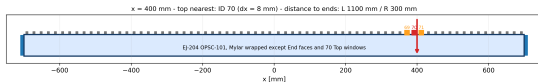
Quantity	Value
End L N_{pe}	92.5 ± 0.6
End R N_{pe}	477.5 ± 2.8
Nearest Top ID	65 (exact geometry)
Nearest Top N_{pe}	408.7 ± 2.3
Nearest Top $\langle t \rangle$	2.936 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=350$ mm



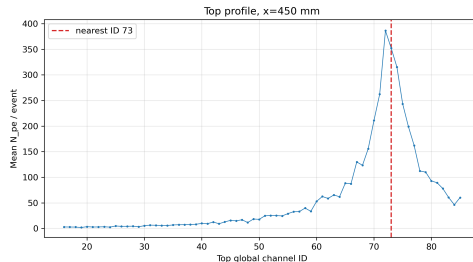
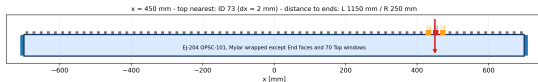
Quantity	Value
End L N_{pe}	84.0 ± 0.5
End R N_{pe}	575.5 ± 3.3
Nearest Top ID	68 (exact geometry)
Nearest Top N_{pe}	349.8 ± 2.0
Nearest Top $\langle t \rangle$	2.777 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=400$ mm



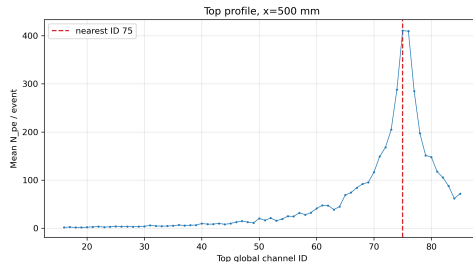
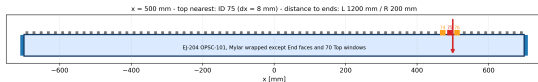
Quantity	Value
End L N_{pe}	75.0 ± 0.5
End R N_{pe}	692.5 ± 4.0
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	411.4 ± 2.4
Nearest Top $\langle t \rangle$	2.943 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=450$ mm



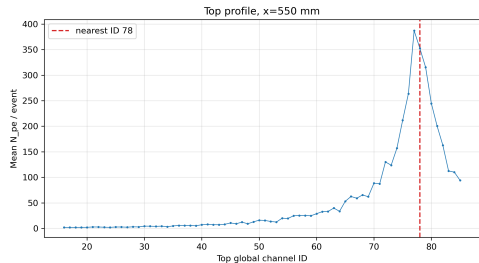
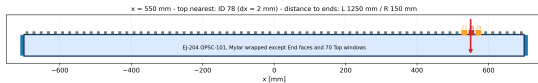
Quantity	Value
End L N_{pe}	67.1 ± 0.4
End R N_{pe}	845.1 ± 4.8
Nearest Top ID	73 (exact geometry)
Nearest Top N_{pe}	352.2 ± 2.1
Nearest Top $\langle t \rangle$	2.774 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=500$ mm



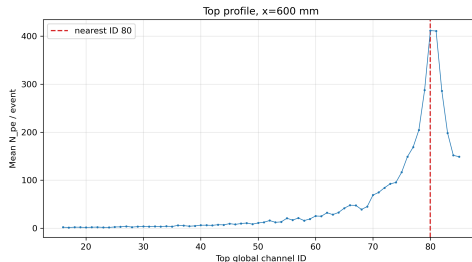
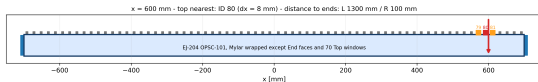
Quantity	Value
End L N_{pe}	59.9 ± 0.4
End R N_{pe}	1039.9 ± 5.9
Nearest Top ID	75 (exact geometry)
Nearest Top N_{pe}	410.5 ± 2.3
Nearest Top $\langle t \rangle$	2.940 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=550$ mm



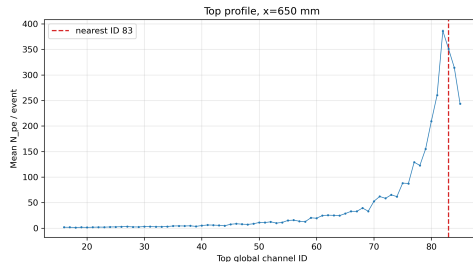
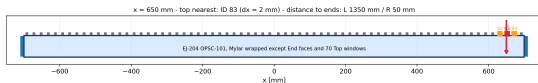
Quantity	Value
End L N_{pe}	55.1 ± 0.4
End R N_{pe}	1325.9 ± 7.7
Nearest Top ID	78 (exact geometry)
Nearest Top N_{pe}	352.3 ± 2.1
Nearest Top $\langle t \rangle$	2.780 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=600$ mm



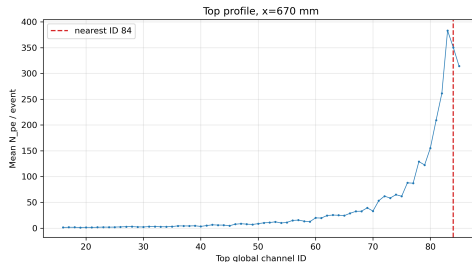
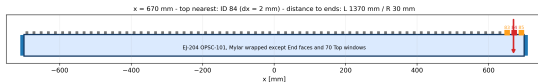
Quantity	Value
End L N_{pe}	48.6 ± 0.3
End R N_{pe}	1687.6 ± 9.3
Nearest Top ID	80 (exact geometry)
Nearest Top N_{pe}	411.6 ± 2.3
Nearest Top $\langle t \rangle$	2.938 ± 0.002 ns
Nearest Top FPT	0.346 ± 0.000 ns

Position $x=650$ mm



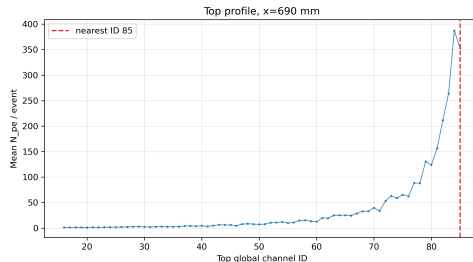
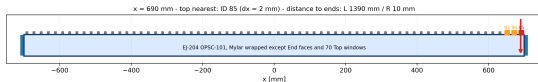
Quantity	Value
End L N_{pe}	44.4 ± 0.3
End R N_{pe}	2357.1 ± 13.0
Nearest Top ID	83 (exact geometry)
Nearest Top N_{pe}	350.9 ± 2.1
Nearest Top $\langle t \rangle$	2.780 ± 0.002 ns
Nearest Top FPT	0.343 ± 0.000 ns

Position $x=670$ mm



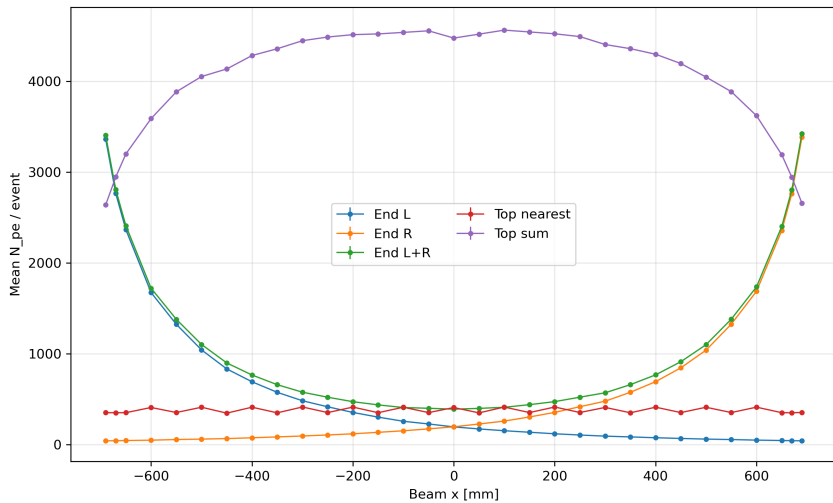
Quantity	Value
End L N_{pe}	42.2 ± 0.3
End R N_{pe}	2763.5 ± 15.3
Nearest Top ID	84 (exact geometry)
Nearest Top N_{pe}	350.2 ± 2.0
Nearest Top $\langle t \rangle$	2.783 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

Position $x=690$ mm

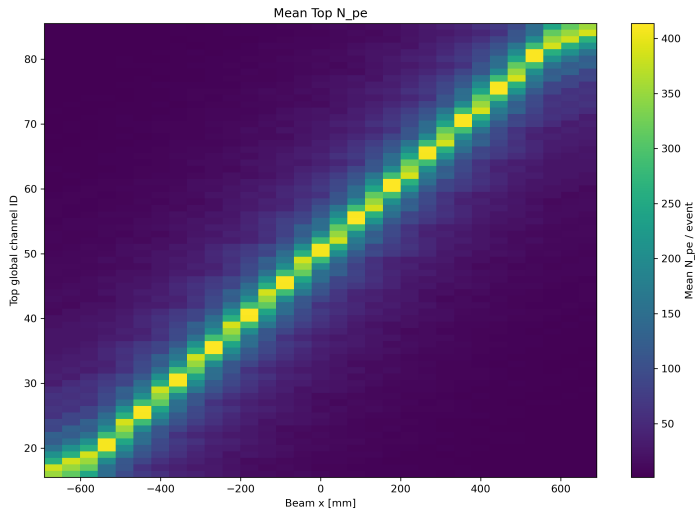


Quantity	Value
End L N_{pe}	41.4 ± 0.3
End R N_{pe}	3383.2 ± 18.3
Nearest Top ID	85 (exact geometry)
Nearest Top N_{pe}	353.0 ± 2.0
Nearest Top $\langle t \rangle$	2.788 ± 0.002 ns
Nearest Top FPT	0.342 ± 0.000 ns

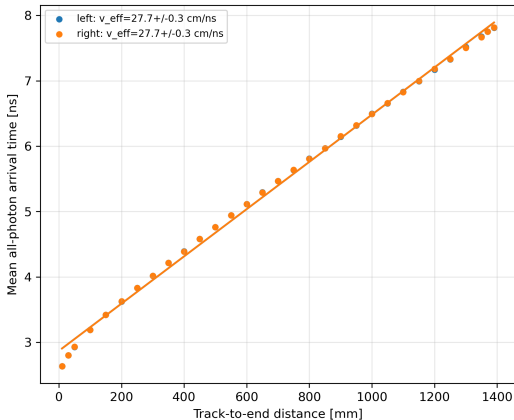
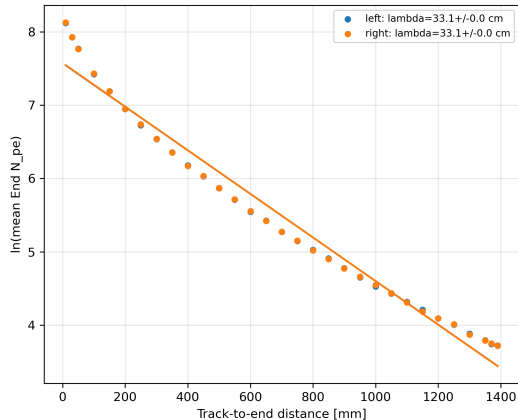
Photon budget versus beam position



Top localization: full coverage diagonal

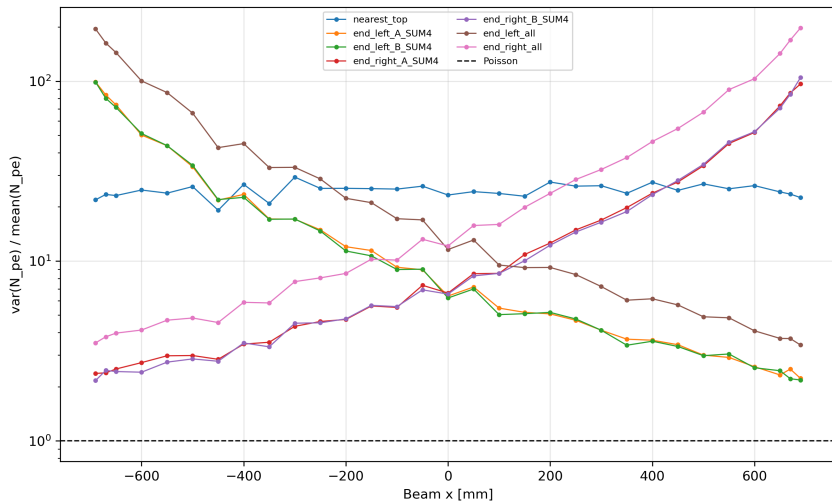


Effective attenuation and propagation fits

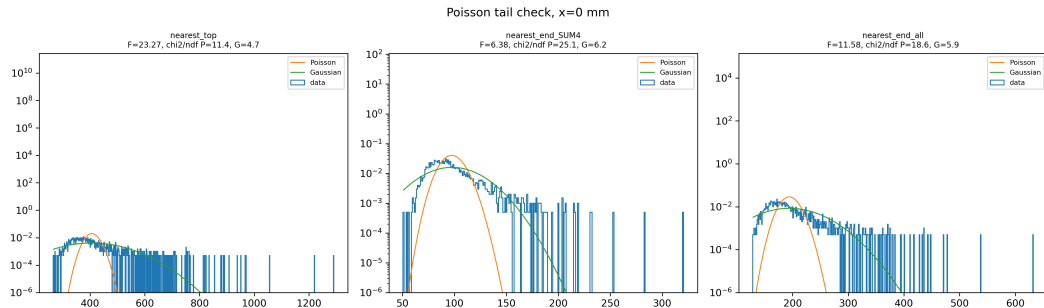


$\lambda_{eff,L} = 33.14 \pm 0.03$ cm, $\lambda_{eff,R} = 33.06 \pm 0.03$ cm; $v_{eff} \simeq 27.67 \pm 0.27$ cm/ns. The 160 cm bulk value and $c/n \simeq 19$ cm/ns are not fit targets.

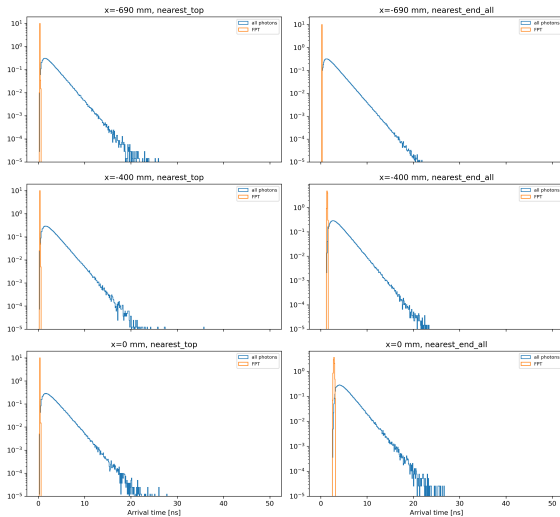
Fano-like variance/mean versus x



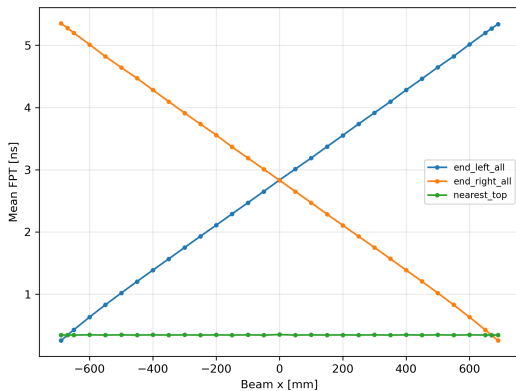
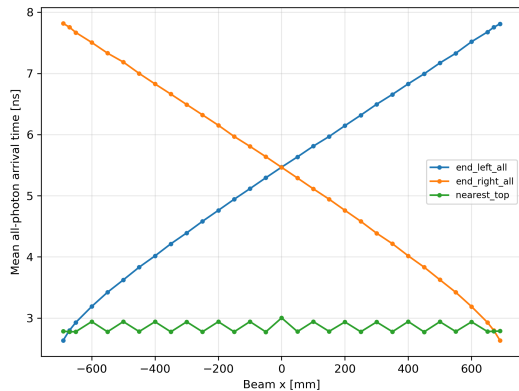
Poisson and Gaussian tail example



Arrival-time examples: all photons and FPT

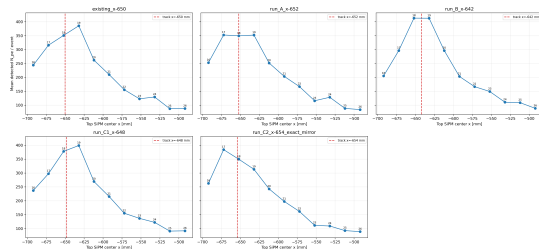


Mean arrival time and FPT versus x



Window-track collection effect: five-run test

Run	contrast [Npe]	significance
Existing -650	34.60 ± 2.94	11.78
A -652 center	2.86 ± 2.65	1.08
B -642 midpoint	-0.33 ± 3.35	-0.10
C1 -648 (+4 mm)	21.08 ± 3.20	6.59
C2 -654 mirror	34.38 ± 2.95	11.64

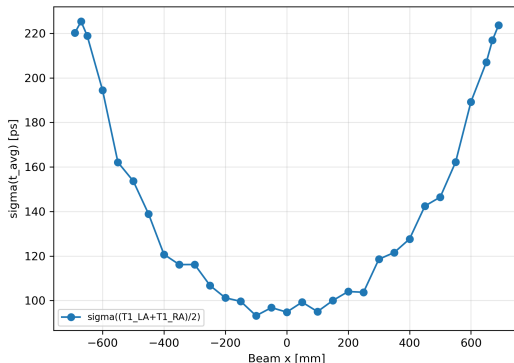
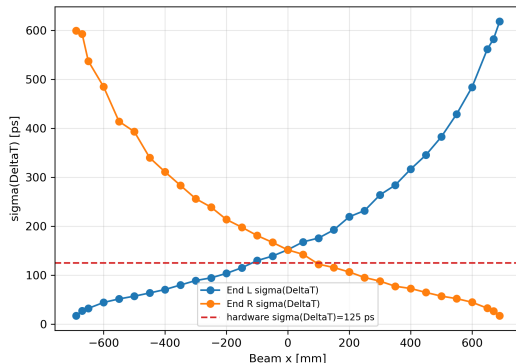


Finding

Local symmetric ± 2 mm effect: about 9.8% at about 12 sigma; null when centered and at midpoint. The scan and channel map remain valid.

Intrinsic End SUM4 timing

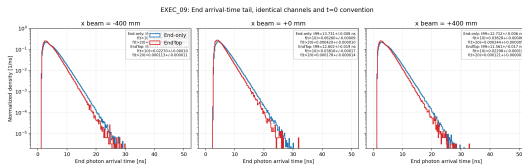
Intrinsic End timing, no SPTR/electronics jitter



$\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$; no SPTR/jitter. The 88 ps hardware context is not directly comparable.

Timing gate lifted: narrower EndTop is physical

x	side	EndTop [ps]	End-only [ps]	ratio
-400	L	50.0 ± 0.8	66.1 ± 0.5	0.756 ± 0.013
-400	R	219.9 ± 3.5	248.4 ± 1.8	0.885 ± 0.015
+0	L	107.5 ± 1.7	131.4 ± 0.9	0.818 ± 0.014
+0	R	107.2 ± 1.7	129.5 ± 0.9	0.828 ± 0.014
+400	L	223.8 ± 3.5	247.5 ± 1.8	0.904 ± 0.016
+400	R	51.5 ± 0.8	65.9 ± 0.5	0.781 ± 0.014



Confirmed mechanism

Across $x = 0, \pm 400$ mm, EndTop lowers t_{99} by 1.14 ns and suppresses late fractions at more than 10.3 sigma.

Top readout: analytic timing estimate only

x [mm]	nearest Top N_{pe}	$\sigma_{t,est}$ [ps]
-690	350.9 ± 2.0	59.92 ± 0.17
-650	351.0 ± 2.0	59.91 ± 0.17
-400	411.3 ± 2.3	55.35 ± 0.16
+0	407.3 ± 2.2	55.62 ± 0.15
+400	411.4 ± 2.4	55.34 ± 0.16
+650	350.9 ± 2.1	59.93 ± 0.18
+690	353.0 ± 2.0	59.75 ± 0.17

Definition and scope

$\sigma_{t,est} = \sqrt{0.7 \text{ ns} \times 1.8 \text{ ns}} / \sqrt{N_{pe}} = 1.122 \text{ ns} / \sqrt{N_{pe}}$. This is an analytic orientation, not a simulated timing resolution; Top has no test-beam counterpart.

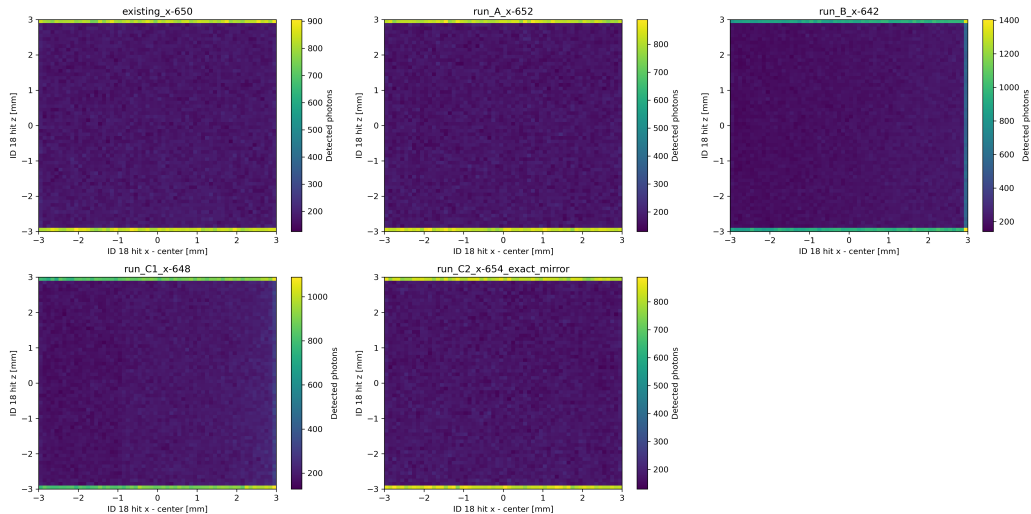
Numerical conclusions

- ▶ At center: End sum 389.6 ± 2.1 , nearest Top 407.3 ± 2.2 ; at $x = -690$ mm: End sum 3404.2 ± 18.3 , nearest Top 350.9 ± 2.0 .
- ▶ λ_{eff} : L 33.14 ± 0.03 cm, R 33.06 ± 0.03 cm; v_{eff} : about 27.67 ± 0.27 cm/ns.
- ▶ Nearest-Top var/mean: median 24.81, range 19.14–29.27; data are strongly over-dispersed relative to Poisson.
- ▶ Intrinsic End σ_{group} : mean 148.3 ps, range 12.1–437.3 ps.
- ▶ EndTop/End-only timing ratio: 0.756 ± 0.016 to 0.904 ± 0.016 .
- ▶ Three findings: window-track dip characterized; late-tail drain confirmed; complete 86-channel coverage and localization gate validated.

Backup: validation and gates

- ▶ 31/31 ROOT files pass: event IDs 0–1999, matching beam x , aggregate IDs 0–85.
- ▶ Top localization passes all positions: strict nearest maximum except the documented $x \equiv 10 \pmod{20}$ window-track effect (maximum among nearest two, deficit below 15%).
- ▶ At $x=300$ mm the two candidates differ by only 0.08 sigma and pass as a statistical tie.
- ▶ MPT checks pass: RINDEX 1.58, yield 10400/MeV, decay 1.8 ns, rise 0.7 ns, ABSLENGTH 160 cm, emission peak 408.8 nm.

Backup: ID18 impact maps



Backup: timing methodology

- ▶ SUM4 groups: {0..3}, {4..7}, {8..11}, {12..15}.
- ▶ Single-PE response: normalized 0.5/5 ns double exponential.
- ▶ Absolute leading-edge threshold: 4 PE equivalent.
- ▶ Estimator: $\sigma(\Delta T_{AB})/\sqrt{2}$.
- ▶ HOOK_WALK: pending; no time-walk correction exists or was invented.
- ▶ EXEC_09 anti-artifact audit verifies common threshold, pulse, $t=0$ and estimator.

Backup: EndTop versus End-only tail audit

EXEC_09: End arrival-time tail, identical channels and t=0 convention

